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REG NO	23MDBCS435
SECTION	A
DEPPT	COMPUTER SCIENCE
SUBJECT	DLD LAB
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LAB TASK : 12

BCD to Seven Segment Display

The Seven segment display is the most frequently used digital display in calculators, digital counters, digital clocks, measuring instruments, etc. Usually, the displays like LED's as well as LCD's are used to display the characters as well as numerical numbers. But, a seven segment display is used to display both the numbers and characters. These displays are frequently driven by the output phases of digital integrated circuits like decade counters as well as latches. A display decoder can be employed for converting BCD code to seven segment code. Generally, it has four input lines as well as seven output lines. This lab discusses how to design a BCD to seven segment display decoder circuit using logic gates.

Truth Table:

A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	0	1	1	0	0	0	0
0	0	1	0	1	1	0	1	1	0	1
0	0	1	1	1	1	1	1	0	0	1
0	1	0	0	0	1	1	0	0	1	1
0	1	0	1	1	0	1	1	0	1	1
0	1	1	0	1	0	1	1	1	1	1
0	1	1	1	1	1	1	0	0	0	0
1	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	1	1	0	1	1

Simplified Boolean Functions:

$$a = A + C + BD + B'D'$$

$$b = B' + CD + C'D'$$

$$c = B + C' + D$$

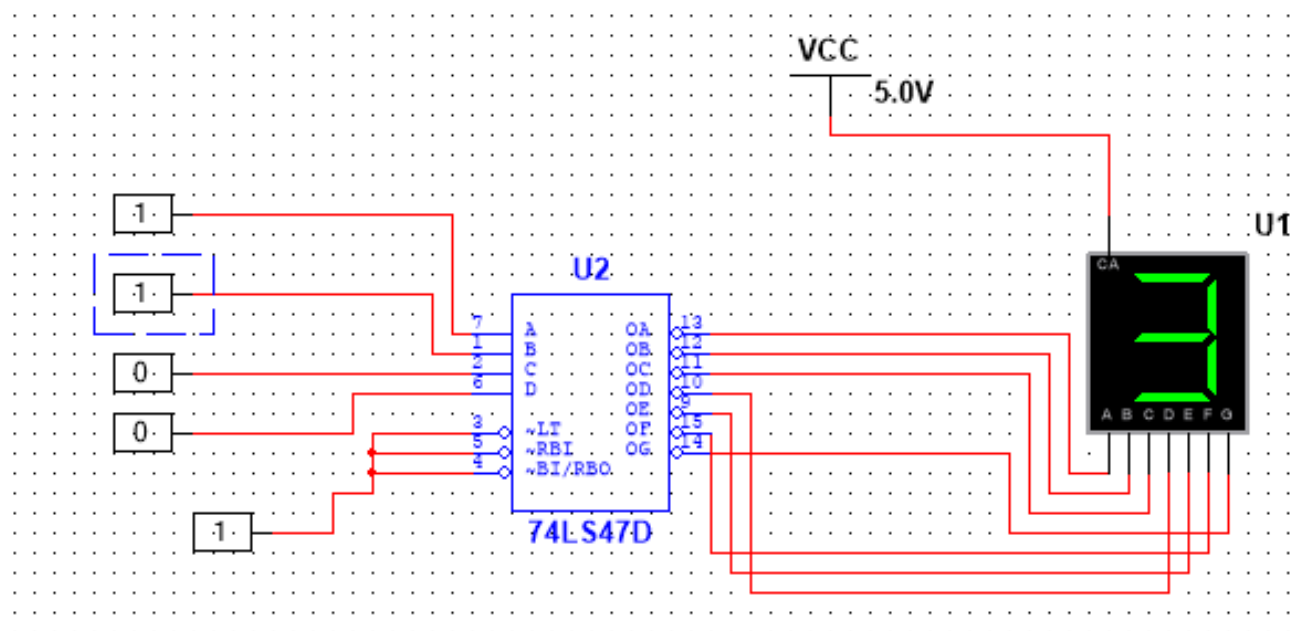
$$d = A + B'C + B'D' + CD' + BC'D$$

$$e = B'D' + CD'$$

$$f = A + BC' + BD' + C'D'$$

$$g = A + BC' + B'C + BD'$$

Circuit Diagram



THE END